GNITS-R- 18 –117FG

G. Narayanamma Institute of Technology & Science

(**Autonomous)** ***(for Women)***

Shaikpet, Hyderabad- 500 104

IV-B.Tech I-Semester *Regular / Supplementary Examinations, Nov-Dec*-2023

DATA SCIENCE USING R

(*Common to CSM & CSD*)

Max. Marks: 70 **Time: 03 Hours**

Note:

1. Question paper comprises of **Part A** and **Part B**.
2. **Part A** is compulsory which carries 10 marks. Answer all questions in Part A.
3. **Part B** (for 60 marks)consists of **five questions** with **“either” “or”** pattern. Each question carries 12 marks and may have a,b,c as sub questions. The student has to answer any one full question.

PART-A

*(Answer 05 questions. Each question carries **2 marks**)*

Q.No.	Question	Marks	CO	BL
Q.1	a) What are the basic data types available in R ?	**[02]**	**CO1**	**[L1]**
	b) Define Missing values and Invalid values and give an example for each.	**[02]**	**CO2**	**[L2]**
	c) What is memorization in Machine Learning?	**[02]**	**CO2**	**[L1]**
	d) Write any two disadvantages of K-Means algorithm.	**[02]**	**CO4**	**[L2]**
	e)What are Non-Seasonal ARIMA models and Seasonal ARIMA models?	**[02]**	**CO5**	**[L1]**

END OF PART A

PART-B

*(Answer **05** full questions. Each question carries **12 marks**)*

Q.No.	Question	Marks	CO	BL
Q.2(a)	How do you install R for data analysis? Describe the sequence of steps.	**[06]**	**CO1**	**[L3]**
(b)	Write short note on the essential steps involved in data collection and management in a Data Science project.	**[06]**	**CO1**	**[L1]**
	OR			
Q.3(a)	Explain Various stages of a Data Science Project.	**[06]**	**CO1**	**[L2]**
(b)	Explain the benefits of using IDEs and text editors in R for data analysis, and provide at least one popular example.	**[06]**	**CO2**	**[L2]**
Q.4(a)	Illustrate the purpose of using the subset() function in R with data frames?	**[06]**	**CO2**	**[L2]**
(b)	Explain the role of data frames in R and how they are structured?	**[06]**	**CO2**	**[L2]**
	OR			
Q.5(a)	Write short note on the purpose of sampling in the context of data modeling and validation?	**[06]**	**CO2**	**[L2]**
(b)	Describe the steps involved in cleaning data in R that include strategies for handling missing values, invalid values.	**[06]**	**CO2**	**[L3]**

Q.6(a)	How do you validate a machine learning model to ensure its reliability?	**[06]**	**CO2**	**[L1]**
(b)	Summarize the common evaluation metrics used for assessing machine learning models.	**[06]**	**CO2**	**[L2]**
	OR			
Q.7(a)	Explain KDD Cup 2009 in machine learning and write short note on their role in model building?	**[06]**	**CO2**	**[L2]**
(b)	Analyze the factors that should be considered while mapping real-world problems to machine learning tasks?	**[06]**	**CO2**	**[L4]**

Q.8(a)	What is the primary objective of linear regression in data analysis?	**[06]**	**CO3**	**[L1]**
(b)	Illustrate the working principle of k-means algorithm with an example.	**[06]**	**CO4**	**[L2]**
	OR			
Q.9(a)	"Hierarchical clustering is advantageous in cluster analysis"- Justify this statement with an example.	**[06]**	**CO4**	**[L5]**
(b)	Outline the key steps of data preparation for clustering analysis?	**[06]**	**CO4**	**[L2]**

Q.10(a)	How can you read time series data in R? Explain with an example.	**[06]**	**CO5**	**[L3]**
(b)	Explain the ARIMA models with the help of a real-world example.	**[06]**	**CO5**	**[L2]**
	OR			
Q.11(a)	Briefly choose the method that is commonly used for making forecasts in time series analysis in the presence of trend and seasonality?	**[06]**	**CO5**	**[L3]**
(b)	Analyze the process of plotting time series data in R.	**[06]**	**CO5**	**[L4]**

END OF PART B

END OF THE QUESTION PAPER